

Reg No.: _____ Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Examination December 2020 (2019 Scheme)

Course Code: MAT201

Course Name: Partial Differential equations and Complex analysis

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks

Marks

- 1 Derive a partial differential equation from the relation $z = (x + y) f(x^2 - y^2)$ (3)
- 2 Solve using direct integration $\frac{\partial^2 u}{\partial x \partial t} = e^{-t} \cos x$ (3)
- 3 Solve $2z = xp + yq$. (3)
- 4 Write any three assumptions in deriving one dimensional heat equation. (3)
- 5 Show that an analytic function $f(z) = u + iv$ is constant if its real part is constant. (3)
- 6 Show that the function $u = \sin x \cos hy$ is harmonic. (3)
- 7 Find the Maclaurin series of $f(z) = \sin z$ (3)
- 8 Evaluate $\oint_C \ln z \, dz$, where C is the unit circle $|z| = 1$. (3)
- 9 Find all singular points and residue of the function $\operatorname{cosec} z$ (3)
- 10 Determine the location and order of zeros of the function $\sin^4 \left(\frac{z}{2}\right)$ (3)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

- 11 (a) Form the Partial differential equation by eliminating the arbitrary constants from $(x - a)^2 + (y - b)^2 = z^2 \cot^2 \alpha$ (5)
- (b) Solve $2xz - p x^2 - 2qxy + pq = 0$ (9)
- 12 (a) Solve $\frac{\partial^3 z}{\partial^2 x \partial y} = \cos(2x + 3y)$ (7)
- (b) Solve $x^2(y - z)p + y^2(z - x)q = z^2(x - y)$ (7)

Module 2

- 13 (a) Derive the solution of the one dimensional wave equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$ using variable separable method. (6)
- (b) An insulated rod of length l has its ends A and B maintained at 0°C and

0800MAT201122001

100° C respectively until steady state conditions prevail. If B is suddenly reduced to 0° C and maintained at 0° C, find the temperature at a distance x from A at time t . (8)

14 (a) Derive the one dimensional heat flow equation. (6)

(b) A tightly stretched string of length l with fixed ends is initially in equilibrium position. If it is set vibrating by giving each points a velocity $v_0 \sin^3(\frac{\pi x}{l})$. Find the displacement $y(x, t)$. (8)

Module 3

15 (a) Find an analytic function whose real part is $u = \sin x \cosh y$ (7)

(b) Find the image of the strip $\frac{1}{2} \leq x \leq 1$ under the transformation $w = z^2$ (7)

16 (a) Check whether $w = \log z$ is analytic. (8)

(b) Show that under the transformation $w = \frac{1}{z}$, the circle $x^2 + y^2 - 6x = 0$ is transformed into a straight line in the W plane. (6)

Module 4

17 (a) Integrate counter clockwise around the unit circle $\oint_C \frac{\sin 2z}{z^4} dz$ (7)

(b) Find the Taylor series of $\frac{1}{1+z}$ about the centre $z_0 = i$ (7)

18 (a) Evaluate $\int_0^{1+i} (x - y + ix^2) dz$ along the parabola $y = x^2$. (7)

(b) Evaluate $\oint_C \frac{\log z}{(z-4)^2} dz$ counter clockwise around the circle $|z - 3| = 2$. (7)

Module 5

19 (a) Find the Laurent's series expansion of $\frac{z^2 - 1}{z^2 - 5z + 6}$ about $z = 0$ in the region (5)

$$2 < |z| < 3$$

(b) Evaluate $\int_0^{2\pi} \frac{d\theta}{\sqrt{2} - \cos \theta}$. (9)

20 (a) Evaluate $\oint_C \frac{z-23}{z^2 - 4z - 5} dz$ where $C : |z - 2 - i| = 3.2$ using Residue theorem. (5)

(b) Evaluate $\int_0^{\infty} \frac{(x^2 + 2) dx}{(x^2 + 1)(x^2 + 4)}$. (9)

KTU 3rd Sem B.tech Degree Exam Dec 2020
(2019 scheme).

Q.P. code: 0800MAT201122001.

MAT201: Partial differential equations
and Complex analysis.

Part A.

1) Derive a partial differential equation
from the relation $z = (x+y)f(x^2-y^2)$

Ans $z = (x+y)f(x^2-y^2)$ ——— ①

$$p = \frac{\partial z}{\partial x} = (x+y)f'(x^2-y^2) \times 2x + f(x^2-y^2) \text{ ——— ②}$$

$$q = \frac{\partial z}{\partial y} = (x+y)f'(x^2-y^2) \times 2y + f(x^2-y^2) \text{ ——— ③}$$

$$\text{①} \Rightarrow f(x^2-y^2) = \frac{z}{x+y}$$

$$\text{②} \Rightarrow p = (x+y)2x f'(x^2-y^2) + \frac{z}{x+y} \text{ ——— ④}$$

$$\text{③} \Rightarrow q = (x+y)2y f'(x^2-y^2) + \frac{z}{x+y} \text{ ——— ⑤}$$

$$\text{④} \Rightarrow f'(x^2-y^2) = \frac{p - \frac{z}{x+y}}{2x(x+y)} \text{ ——— ⑥}$$

$$\text{⑤} \Rightarrow f'(x^2-y^2) = \frac{q - \frac{z}{x+y}}{-2y(x+y)} \text{ ——— ⑦}$$

$$\text{From ⑥, ⑦} \Rightarrow \frac{p - \frac{z}{x+y}}{2x(x+y)} = \frac{q - \frac{z}{x+y}}{-2y(x+y)} \Rightarrow \underline{\underline{py + qx - z = 0}}$$

2) Solve using direct integration $\frac{\partial^2 y}{\partial x \partial t} = e^{-t} \cos x$

Ans $\frac{\partial^2 y}{\partial x \partial t} = e^{-t} \cos x$

Integrating partially w.r. to x

$$\frac{\partial y}{\partial t} = e^{-t} \sin x + f(t).$$

Integrating partially w.r. to t

$$u = -e^{-t} \sin x + \int f(t) dt + g(x)$$

3) Solve $2z = xp + yq$.

Ans $Pp + Qq = R$ form.

$$xp + yq = 2z$$

$$AE \Rightarrow \frac{dx}{p} = \frac{dy}{q} = \frac{dz}{R}$$

$$\Rightarrow \frac{dx}{x} = \frac{dy}{y} = \frac{dz}{2z} \quad \text{--- (1)}$$

Solving the subsidiary equations and finding two sets of solutions

$$\frac{dx}{x} = \frac{dy}{y} \Rightarrow \int \frac{dx}{x} = \int \frac{dy}{y} \Rightarrow \log x = \log y + \log c_1$$

$$\Rightarrow \log c_1 = \log x - \log y$$

$$\Rightarrow c_1 = \frac{x}{y} \quad (\text{1st soln}).$$

$$\frac{dy}{y} = \frac{dz}{2z} \Rightarrow 2 \log y = \frac{1}{2} \log z + \log c_2$$

$$\Rightarrow \log c_2 = \log y - \frac{1}{2} \log z = \log y - \log z^{\frac{1}{2}}$$

$$\Rightarrow \log c_2 = \log \left(\frac{y}{z^{\frac{1}{2}}} \right) \Rightarrow c_2 = \frac{y}{z^{\frac{1}{2}}} \quad (\text{or}) \quad \frac{y^2}{z}$$

∴ soln is $\phi\left(\frac{x}{y}, \frac{y}{z}\right) = 0$ or $\phi\left(\frac{x}{y}, \frac{y^2}{z}\right) = 0$

4) Write any 3 assumptions in deriving one dimensional heat equation.

Ans

- 1) The sides of the bar are insulated so that there is no radiation from the sides.
- 2) It is assumed that the bar is raised to an assigned temperature distribution at time $t=0$ and then heat is allowed to flow by conduction.
- 3) The temp at all pts of any cross section is same.

5) Show that an analytic function $f(z) = u + iv$ is constant if its real part is constant.

Ans

$f(z) = u + iv$ — (1)

given Real part is constant $\Rightarrow u = \text{constant}$

$\Rightarrow \frac{\partial u}{\partial x} = 0, \frac{\partial u}{\partial y} = 0$

By CR equations $\left. \begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} &= -\frac{\partial v}{\partial x} \end{aligned} \right\} \quad \text{--- (1)}$

CR apply in (1) $\Rightarrow \therefore \frac{\partial v}{\partial y} = 0, \frac{\partial v}{\partial x} = 0$

Integrate

∴ $v = \text{constant}$

∴ $f(z)$ is constant

6) Show that the function $u = \sin x \cosh y$ is harmonic.

Ans $u = \sin x \cosh y$

$$u_x = \cos x \cosh y$$

$$u_{xx} = -\sin x \cosh y$$

$$u_y = \sin x \sinh y$$

$$u_{yy} = \sin x \cosh y$$

$$\therefore u_{xx} + u_{yy} = 0 \quad \therefore u \text{ satisfy Laplace equ.}$$

$\therefore u$ is harmonic

7) Find the Maclaurin Series of $f(z) = \sin z$.

Ans $f(z) = \sin z$

$$f(0) = 0$$

$$f'(z) = \cos z$$

$$f'(0) = 1$$

$$f''(z) = -\sin z$$

$$f''(0) = -0$$

$$f'''(z) = -\cos z$$

$$f'''(0) = -1$$

$$\therefore f(z) = f(0) + \frac{z}{1!} f'(0) + \frac{z^2}{2!} f''(0) + \frac{z^3}{3!} f'''(0) + \dots$$

$$= \frac{z}{1!} - \frac{z^3}{3!} + \dots$$

8) Evaluate $\oint_C \ln z \, dz$, where C is the unit circle $|z|=1$.

Ans $C: |z|=1 \Rightarrow z = re^{i\theta}$ (polar form)

here
 $z = e^{i\theta}$

$\ln z$ is not analytic

$$dz = i e^{i\theta} d\theta$$

$\theta \rightarrow 0 \text{ to } 2\pi$

$$\oint_C \ln z \, dz = \int_0^{2\pi} i(\theta + 2\pi n) (i e^{i\theta} d\theta)$$

$$= i \int_0^{2\pi} (\theta + 2\pi n) (e^{i\theta}) d\theta$$

$$= - \int_0^{2\pi} (\theta + 2\pi n) e^{i\theta} d\theta = - \left[(\theta + 2\pi n) \frac{e^{i\theta}}{i} + e^{i\theta} \right]_0^{2\pi}$$

$$= \underline{\underline{2\pi i}}$$

$$\ln e^{i\theta} = i\theta$$

$$\ln z = i(\theta + 2\pi n) \text{ for whole } n$$

9) Find all singular points and residue of the function $\operatorname{cosec} z$.

Ans

$$f(z) = \operatorname{cosec} z = \frac{1}{\sin z}$$

$$= \frac{1}{z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots}$$

$$= \frac{1}{z} + \frac{z}{3!} - \dots$$

$$= \frac{1}{z} \left[1 - \left(\frac{z^2}{3!} + \frac{z^4}{5!} - \dots \right) \right]$$

$z=0$ is a pole.
Res at $z=0$ is 1.

Part B

11. a) Form a Partial differential equation by eliminating the arbitrary constants from
 $(x-a)^2 + (y-b)^2 = z^2 \cot^2 \alpha$.

Ans $(x-a)^2 + (y-b)^2 = z^2 \cot^2 \alpha$ — (1) $\left\{ \begin{array}{l} \frac{\partial z}{\partial x} = p \\ \frac{\partial z}{\partial y} = q \end{array} \right\}$

diff. w.r. to x

$$2(x-a) + 0 = 2z \frac{\partial z}{\partial x} \cot^2 \alpha \Rightarrow (x-a) = z p \cot^2 \alpha \quad \text{--- (2)}$$

diff. w.r. to y

$$2(y-b) = 2z \frac{\partial z}{\partial y} \cot^2 \alpha \Rightarrow (y-b) = z q \cot^2 \alpha \quad \text{--- (3)}$$

substitute (2), (3) in eqn (1).

$$\therefore (1) \Rightarrow (p^2 + q^2) z^2 \cot^4 \alpha = z^2 \cot^2 \alpha$$

$$\Rightarrow \underline{\underline{p^2 + q^2 = \tan^2 \alpha}}$$

b) Solve $2xz - px^2 - 2qxy + pq = 0$.

Ans $f(x, y, z, p, q) = 0 \Rightarrow 2xz - px^2 - 2qxy + pq = 0$ — (1)

$$\therefore f = 2xz - px^2 - 2qxy + pq \quad \text{--- (2)}$$

Charpit's subsidiary eqn are

$$\frac{dx}{-f_p} = \frac{dy}{-f_q} = \frac{dz}{-pf_p - qf_q} = \frac{dp}{f_x + pf_z} = \frac{dq}{f_y + qf_z} \quad \text{(A)}$$

$$\frac{dx}{-(x^2 + q)} = \frac{dy}{-(2xy + p)} = \frac{dz}{-p(-x^2 + q) - 2(-2xy + p)} = \frac{dp}{2z - 2q} = \frac{dq}{0}$$

$$\begin{array}{ccccccc} \frac{dx}{x^2-z} & = & \frac{dy}{2xy-p} & = & \frac{dz}{x^2p-2pz+2xyq} & = & \frac{dp}{2z-2qy} = \frac{dq}{0} \\ \text{I} & & \text{II} & & \text{III} & & \text{IV} & & \text{V} \end{array}$$

From Vth $\Rightarrow dq=0 \Rightarrow \underline{q=a}$ substitute in equ^①

$$\therefore \text{①} \Rightarrow 2xz - px^2 - 2axy + pa = 0$$

$$\Rightarrow 2x(z-ay) = p(x^2-a)$$

$$\Rightarrow p = \frac{2x(z-ay)}{(x^2-a)}$$

Substitute the values p, q in equ

$$dz = p dx + q dy \Rightarrow dz = \frac{2x(z-ay)}{(x^2-a)} dx + a dy$$

$$\Rightarrow dz - a dy = \frac{2x(z-ay)}{(x^2-a)} dx$$

$$\Rightarrow \int \frac{dz - a dy}{z - ay} = \int \frac{2x}{x^2-a} dx$$

$$\Rightarrow \log(z-ay) = \log(x^2-a) + \log b$$

$$\Rightarrow z-ay = b(x^2-a)$$

$$\Rightarrow \underline{\underline{z = ay + b(x^2-a)}}$$

12. a) Solve $\frac{\partial^3 z}{\partial^2 x \partial y} = \cos(2x+3y)$

Ans

$$\frac{\partial^3 z}{\partial^2 x \partial y} = \cos(2x+3y)$$

Integrating Partially w.r. to x .

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\sin(2x+3y)}{2} + u(y), \text{ where } u(y) \text{ is a fun. of } y \text{ only}$$

Integrating Partially w.r. to x

$$\frac{\partial z}{\partial y} = -\frac{\cos(2x+3y)}{4} + x u(y) + v(y), \text{ where } v(y) \text{ is fun. of } y \text{ only}$$

Integrating partially w.r. to y

$$z = -\frac{\sin(2x+3y)}{12} + x \int u(y) dy + \int v(y) dy + w(x), \text{ where } w(x) \text{ is fun. of } x \text{ only}$$

b) solve $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$.

Ans: $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$ ①

$$AE \Rightarrow \frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)}$$

choose $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$.

$$\Rightarrow \frac{\frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz}{x(y-z) + y(z-x) + z(x-y)}$$

$$\Rightarrow \frac{\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z}}{0}$$

$$\Rightarrow \frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0$$

$$\Rightarrow \log x + \log y + \log z = \log c_1$$

$$\Rightarrow \underline{\underline{c_1 = xyz}}$$

$$\text{choose } \frac{1}{x^2}, \frac{1}{y^2}, \frac{1}{z^2}.$$

$$\Rightarrow \frac{\frac{1}{x^2} dx + \frac{1}{y^2} dy + \frac{1}{z^2} dz}{y-z + z-x + x-y}$$

$$\Rightarrow \frac{\frac{dx}{x^2} + \frac{dy}{y^2} + \frac{dz}{z^2}}{0}$$

$$\Rightarrow \frac{dx}{x^2} + \frac{dy}{y^2} + \frac{dz}{z^2} = 0 \Rightarrow \underline{\underline{-\frac{1}{x} - \frac{1}{y} - \frac{1}{z} = c_2}}$$

$$\Rightarrow \text{G.S is } \phi(xyz, \frac{1}{x} + \frac{1}{y} + \frac{1}{z}) = 0$$

13. a) Derive the solution of the one dimensional wave equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$ using variable separable method.

Ans

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \text{--- (1)}$$

using variable separable method

Let soln of (1) be $y = XT$ --- (2)

X is fun of x alone

T is fun of t alone

$$\left| \begin{array}{l} \frac{\partial y}{\partial x} = X' T \\ \frac{\partial^2 y}{\partial x^2} = X'' T \end{array} \right| \left| \begin{array}{l} \frac{\partial y}{\partial t} = X T' \\ \frac{\partial^2 y}{\partial t^2} = X T'' \end{array} \right|$$

$$(1) \Rightarrow X T'' = c^2 X'' T$$

$$\Rightarrow \frac{X''}{X} = \frac{1}{c^2} \frac{T''}{T}$$

$$\Rightarrow \frac{X''}{X} = \frac{1}{c^2} \frac{T''}{T} = K$$

I II III

$$\frac{X''}{X} = K$$

$$X'' - XK = 0 \quad \text{--- (3)}$$

$$\frac{1}{c^2} \frac{T''}{T} = K$$

$$T'' - c^2 K T = 0 \quad \text{--- (4)}$$

Solve (3) and (4)

i) when k is +ve i.e. $k = p^2$

Substitute in (3), (4)

$$(3) \Rightarrow x'' - x p^2 = 0$$

$$AE \Rightarrow m^2 - p^2 = 0$$

$$\Rightarrow m = \pm p \text{ (distinct real root)}$$

$$\Rightarrow \underline{x = c_1 e^{px} + c_2 e^{-px}}$$

$$(4) \Rightarrow T'' - c^2 p^2 T = 0$$

$$AE \Rightarrow (m^2 - c^2 p^2) T = 0$$

$$\Rightarrow m = \pm cp \text{ (distinct)}$$

$$\Rightarrow \underline{T = c_3 e^{cpt} + c_4 e^{-cpt}}$$

$$\text{if } k = p^2 \Rightarrow \text{soln is } \underline{y = (c_1 e^{px} + c_2 e^{-px})(c_3 e^{cpt} + c_4 e^{-cpt})}$$

ii) when k is -ve i.e. $k = -p^2$

Substitute in (3), (4)

$$(3) \Rightarrow x'' + x p^2 = 0$$

$$AE \Rightarrow (m^2 + p^2) x = 0$$

$$\Rightarrow m = \pm i p \text{ (complex)}$$

$$\Rightarrow \underline{x = c_5 \cos px + c_6 \sin px}$$

$$(4) T'' + c^2 p^2 T = 0$$

$$AE \Rightarrow (m^2 + c^2 p^2) T = 0$$

$$\Rightarrow m = \pm i cp \text{ (complex)}$$

$$\Rightarrow \underline{T = c_7 \cos cpt + c_8 \sin cpt}$$

$$\text{if } k = -p^2 \Rightarrow \text{soln is } \underline{y = [c_5 \cos px + c_6 \sin px][c_7 \cos cpt + c_8 \sin cpt]}$$

i.e. when $k=0$

substitute in (3), (4)

$$(3) \Rightarrow x'' = 0$$

$$(4) \Rightarrow T'' = 0$$

$$AE \Rightarrow m^2 x = 0$$

$$AE \Rightarrow m^2 T = 0$$

$$\Rightarrow m = 0, 0 \text{ (equal)}$$

$$\Rightarrow T = 0, 0 \text{ (equal)}$$

$$\Rightarrow X = [c_9 x + c_{10}]$$

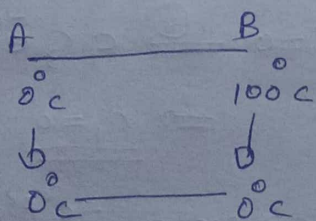
$$\Rightarrow T = [c_{11} t + c_{12}]$$

∴ of $k=0 \Rightarrow$ soln is $y = [c_9 x + c_{10}][c_{11} t + c_{12}]$

b) An insulated rod of length l has its ends A and B maintained 0°C and 100°C respectively until steady state conditions prevail. If B is suddenly reduced to 0°C and maintained at 0°C find the temp at a distance x from A at time t .

Ans Equ of heat conduction is $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ ①

Initial condition $u(x, 0) = \frac{100x}{l}$



Boundary Condition : $u(0,t) = 0$ — (2)
 $u(l,t) = 100$ — (3)

In steady state $\Rightarrow \frac{\partial u}{\partial t} = 0$ in (1)

(1) $\Rightarrow \frac{\partial^2 u}{\partial x^2} = 0$ (Integrate)

$\Rightarrow \frac{\partial u}{\partial x} = A$ (Integrate)

$\Rightarrow u = Ax + B$

$\Rightarrow u(x,t) = Ax + B$ — (4)

apply BC in (4) $\Rightarrow u(0,t) = 0$ in (4) $\underline{\underline{B = 0}}$

$u(l,t) = 100$ in (4) $\underline{\underline{A = \frac{100}{l}}}$

$\therefore$ (4) $\Rightarrow \underline{\underline{u(x,t) = \frac{100x}{l}}}$

Take $f(x) = \frac{100x}{l}$ is initial temp.

at the end B is reduced to 0°C and maintained at 0°C , the boundary conditions for the subsequent flow are

$u(0,t) = 0$ — (6)

$u(l,t) = 0$ — (7)

$u(x,0) = \frac{100x}{l}$ — (5) is the IC

Soln of eqn (1) is $u(x,t) = (c_1 \cos px + c_2 \sin px) e^{-c^2 p^2 t}$ — (8)

Apply Bc (6), (7) in (8) $\Rightarrow u(0, t) = 0 \xrightarrow{\text{in (8)}} c_1 c_3 e^{-c^2 p^2 t} = 0$

$$\Rightarrow \underline{c_1 = 0}$$

substitute in (8) $\Rightarrow u(x, t) = c_2 \sin px \cdot c_3 e^{-c^2 p^2 t}$ — (9)

apply $u(l, t) = 0 \xrightarrow{\text{in (9)}} c_2 \sin pl \cdot c_3 e^{-c^2 p^2 t} = 0$

$$\Rightarrow \sin pl = 0$$

$$\Rightarrow p = \frac{n\pi}{l}, \quad n = 1, 2, \dots$$

substitute in (9) $\Rightarrow u(x, t) = c_2 c_3 \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{c^2 n^2 \pi^2 t}{l^2}}$

$$\Rightarrow u(x, t) = b_n \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{c^2 n^2 \pi^2 t}{l^2}},$$

where $c_2 c_3 = b_n$

Take summation

$$\Rightarrow u(x, t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{c^2 n^2 \pi^2 t}{l^2}}$$

— (10)

apply IC (5) in (10) $\Rightarrow u(x, 0) = \frac{100x}{l} \xrightarrow{\text{in (10)}} \frac{100x}{l} = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$ is

the F. sine series, where

$$b_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2}{l} \int_0^l \frac{100x}{l} \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{200}{l^2} \int_0^l \underbrace{x}_u \cdot \underbrace{\sin\left(\frac{n\pi x}{l}\right)}_v dx$$

$$= \frac{200}{l^2} \left[-x \frac{l}{n\pi} \cos\left(\frac{n\pi x}{l}\right) + \frac{l^2}{n^2 \pi^2} \sin\left(\frac{n\pi x}{l}\right) \right]_0^l$$

$$= \frac{200(-1)^{n+1}}{n\pi}$$

soln is

substitute in (10) $\Rightarrow u(x, t) = \sum_{n=1}^{\infty} \frac{200}{n\pi} (-1)^{n+1} \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{c^2 n^2 \pi^2 t}{l^2}}$

14) a) Derive the one dimensional heat flow equation.

Ans

Let R_1 be quantity of heat flowing into the section at a distance x in time δt then

$$R_1 = -k A \left(\frac{\partial u}{\partial x} \right)_x \delta t$$

Let R_2 be quantity of heat flowing into the section at a distance $x + \delta x$ in time δt then

$$R_2 = -k A \left(\frac{\partial u}{\partial x} \right)_{x+\delta x} \delta t$$

$$R_1 - R_2 = k A \delta t \left[\left(\frac{\partial u}{\partial x} \right)_{x+\delta x} - \left(\frac{\partial u}{\partial x} \right)_x \right] \quad \text{--- (1)}$$

$$\text{mass} = (\rho A \delta x) \delta u \quad \text{--- (2)}$$

$$\text{①} = \text{②} \Rightarrow \frac{k}{\rho \sigma} \left[\frac{\left(\frac{\partial u}{\partial x} \right)_{x+\delta x} - \left(\frac{\partial u}{\partial x} \right)_x}{\delta x} \right] = \frac{\partial u}{\partial t}$$

$$\Rightarrow \frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} \right), \text{ where } c^2 = \frac{k}{\rho \sigma}$$

is the one dimensional heat equation

b) A tightly stretched string of length l with fixed ends is initially in equilibrium position. If it is set vibrating by giving each points a velocity $y_0 \sin^3\left(\frac{\pi x}{l}\right)$. Find the

displacement $y(x, t)$.

Ans The equ. of the vibrating string is

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \text{--- (1)}$$

$$BC \Rightarrow y(0, t) = 0 \quad \text{--- (2)}$$

$$y(l, t) = 0 \quad \text{--- (3)}$$

$$IC \Rightarrow \left(\frac{\partial y}{\partial t} \right)_{t=0} = 0 \quad \text{--- (4)}$$

$$y(x, 0) = y_0 \sin^3\left(\frac{\pi x}{l}\right) = f(x) \quad \text{--- (5)}$$

$$\text{soln of (1) is } y(x, t) = [c_1 \cos px + c_2 \sin px] [c_3 \cos cpt + c_4 \sin cpt] \quad \text{--- (6)}$$

$$\text{apply BC} \Rightarrow y(0, t) = 0 \xrightarrow{\text{in (6)}} 0 = c_1 [c_3 \cos cpt + c_4 \sin cpt] \\ \Rightarrow \underline{c_1 = 0}$$

$$\text{substitute in (6)} \Rightarrow y(x, t) = c_2 \sin px [c_3 \cos cpt + c_4 \sin cpt] \quad \text{--- (7)}$$

$$\text{apply IC} \Rightarrow \left(\frac{\partial y}{\partial t} \right)_{t=0} = 0 \xrightarrow{\text{in (7)}} 0 = c_2 \sin px c_4 cp$$

$$\xrightarrow{\text{in (7)}} \sin \omega \frac{\partial y}{\partial t} = c_2 \sin px [-cp c_3 \sin cpt + cp c_4 \cos cpt] \Rightarrow \underline{c_4 = 0}$$

$$\text{substitute in (7)} \Rightarrow y(x, t) = c_2 c_3 \sin px \cos cpt \quad \text{--- (8)}$$

$$\text{apply BC} \Rightarrow y(l, t) = 0 \xrightarrow{\text{in (8)}} \sin pl = 0 \\ \Rightarrow p = \frac{n\pi}{l}, n = 1, 2, \dots$$

$$\text{Substitute in (8)} \Rightarrow y(x, t) = c_2 c_3 \sin\left(\frac{n\pi x}{l}\right) \cos\left(\frac{n\pi ct}{l}\right)$$

$$\Rightarrow y(x, t) = b_n \sin\left(\frac{n\pi x}{2}\right) \cos\left(\frac{n\pi ct}{2}\right), \text{ where } c_2 c_3 = b_n$$

$n=1, 2, \dots$

Take summation

$$y(x, t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{2}\right) \cos\left(\frac{n\pi ct}{2}\right) \quad \text{--- (9)}$$

$$\text{Apply IC} \Rightarrow y(x, 0) = y_0 \sin^3\left(\frac{\pi x}{2}\right) \xrightarrow{\text{in (9)}} y_0 \sin^3\left(\frac{\pi x}{2}\right) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{2}\right)$$

$$\Rightarrow y_0 \left[\frac{3 \sin\left(\frac{\pi x}{2}\right) - \sin\left(\frac{3\pi x}{2}\right)}{4} \right] = b_1 \sin\left(\frac{\pi x}{2}\right) + b_2 \sin\left(\frac{2\pi x}{2}\right) + \dots$$

$$\Rightarrow \frac{3y_0}{4} \sin\left(\frac{\pi x}{2}\right) - \frac{y_0}{4} \sin\left(\frac{3\pi x}{2}\right) = b_1 \sin\left(\frac{\pi x}{2}\right) + b_2 \sin\left(\frac{2\pi x}{2}\right) + b_3 \sin\left(\frac{3\pi x}{2}\right) + \dots$$

equating $\sin\left(\frac{\pi x}{2}\right), \sin\left(\frac{3\pi x}{2}\right)$ on both side

$$\underline{\underline{b_1 = \frac{3y_0}{4}}} \quad \underline{\underline{b_3 = -\frac{y_0}{4}}} \quad b_2 = b_4 = b_5 = b_6 = b_7 = \dots = 0$$

$$\text{Substitute in (9)} \Rightarrow y(x, t) = \frac{3y_0}{4} \sin\left(\frac{\pi x}{2}\right) \cos\left(\frac{\pi ct}{2}\right) - \frac{y_0}{4} \sin\left(\frac{3\pi x}{2}\right) \cos\left(\frac{3\pi ct}{2}\right)$$

15) a) Find an analytic function whose real part is $u = \sin x \cosh y$.

Ans $u = \sin x \cosh y$

$$u_x = \cos x \cosh y \quad \text{--- (1)} \quad u_y = \sin x \sinh y \quad \text{--- (2)}$$

$$u_{xx} = -\sin x \cosh y \quad u_{yy} = \sin x \cosh y$$

$$\therefore u_{xx} + u_{yy} = 0$$

$\therefore u$ is harmonic

To find harmonic conjugate 'v'

$$u = \sin x \cosh y$$

$$\text{apply CR} \Rightarrow \left. \begin{array}{l} u_x = v_y \\ u_y = -v_x \end{array} \right\} \text{--- (3)}$$

$$\text{apply CR in (2)} \Rightarrow u_y = -v_x \xrightarrow{\text{in (2)}} -v_x = \sin x \sinh y$$

$$\Rightarrow v_x = -\sin x \sinh y$$

Integrate w.r. to x

$$\Rightarrow v = -\sinh y \cos x + h(y) \text{--- (4)}$$

partially diff w.r. to y

$$\Rightarrow v_y = \cosh y \cos x + h'(y)$$

apply CR ($u_x = v_y$)

$$\Rightarrow u_x = \cosh y \cos x + h'(y)$$

from (1)

$$\Rightarrow \cos x \cosh y = \cosh y \cos x + h'(y)$$

$$\Rightarrow h'(y) = 0$$

Integrate w.r. to y

$$\Rightarrow \underline{h(y) = C} \text{ substitute in (4)}$$

$$\text{(4)} \Rightarrow \underline{v = -\sinh y \cos x + C}$$

To find analytic function i.e. $f(z)$

$$f'(z) = u_x + i v_x$$

$$= \cos x \cosh y + i(-y)$$

$$f'(z) = \cos x \cosh y + i(-\sin x \sinh y)$$

$$\left. \begin{array}{l} \text{put } x=z \\ y=0 \end{array} \right\}$$

$$\therefore f'(z) = \cos z$$

Integrate

$$f(z) = \sin z + C$$

b) Find the image of the strip $\frac{1}{2} \leq x \leq 1$ under the transformation $w = z^2$.

Ans. $w = z^2$

$$u + iv = (x + iy)^2$$

$$= (x^2 - y^2) + i(2xy)$$

$$u = x^2 - y^2 \quad \text{--- (1)} \quad v = 2xy \quad \text{--- (2)}$$

$\frac{1}{2} \leq x \leq 1 \Rightarrow$ lines b/w $x = \frac{1}{2}$ and $x = 1$.

$$x = \frac{1}{2} \text{ in (1)} \Rightarrow u = \frac{1}{4} - y^2 \quad \text{--- (3)} \Rightarrow y^2 = \frac{1}{4} - u$$

$$(2) \Rightarrow v = 2y \quad \text{--- (4)}$$

$$(4)^2 \Rightarrow v^2 = 4y^2$$

$$v^2 = \frac{1}{4} - u \text{ which is a parabola}$$

$$\text{when } x=1 \Rightarrow \text{in } \textcircled{1} \Rightarrow u = 1 - y^2 \text{ --- } \textcircled{5} \Rightarrow y^2 = 1 - u.$$

$$\textcircled{2} \Rightarrow v = 2y. \text{ --- } \textcircled{6}$$

$$\textcircled{6}^2 \Rightarrow (v)^2 = 4y^2$$

$$\underline{\underline{v^2 = 4(1-u)}} \text{ which is a parabola}$$

Hence image of the strip $\frac{1}{2} \leq x \leq 1$ is the region b/w the parabolas $v^2 = \frac{1}{4} - u$ and $v^2 = 4(1-u)$

16. a) check whether $w = \log z$ is analytic.

Ans $w = \log z$

$$u + iv = \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1}\left(\frac{y}{x}\right)$$

$$u = \frac{1}{2} \log(x^2 + y^2)$$

$$v = \tan^{-1}(y/x)$$

$$u_x = \frac{1}{2} \times \frac{1}{x^2 + y^2} \times 2x = \frac{x}{x^2 + y^2}$$

$$v_x = \frac{1}{1 + \frac{y^2}{x^2}} \times \frac{-y}{x^2} = \frac{-y}{x^2 + y^2}$$

$$u_y = \frac{1}{2} \times \frac{1}{x^2 + y^2} \times 2y = \frac{y}{x^2 + y^2}$$

$$v_y = \frac{1}{1 + \frac{y^2}{x^2}} \times \frac{1}{x} = \frac{x}{x^2 + y^2}$$

$$\therefore \text{CR equ are satisfied } u_x = v_y$$

$$u_y = -v_x$$

and partial derivatives are continuous.

$\therefore f(z)$ is analytic

b). show that under the transformation $w = \frac{1}{z}$ the circle $x^2 + y^2 - 6x = 0$ is transformed into a straight line in the w plane.

Ans

$$w = \frac{1}{z} \Rightarrow z = \frac{1}{w} \Rightarrow x + iy = \frac{1}{u + iv}$$

$$\left\{ \begin{aligned} u + iv &= \frac{1}{x + iy} \end{aligned} \right\}$$

$$\Rightarrow x + iy = \frac{u - iv}{u^2 + v^2}$$

$$\Rightarrow x = \frac{u}{u^2 + v^2}, \quad y = \frac{-v}{u^2 + v^2}$$

$$\text{given } x^2 + y^2 - 6x = 0 \Rightarrow \left(\frac{u}{u^2 + v^2} \right)^2 + \left(\frac{-v}{u^2 + v^2} \right)^2 - 6 \frac{u}{u^2 + v^2} = 0$$

$$\Rightarrow \underline{\underline{u = \frac{1}{6}}}$$

(after simplification)
which is a st. line
in w plane

17. a) Integrate counter clockwise around the unit circle $\oint_C \frac{\sin 2z}{z^4} dz$.

Ans

$$\oint_C \frac{\sin 2z}{z^4} dz = \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz = \frac{2\pi i}{n!} f^{(n)}(z_0)$$

$$f(z) = \sin 2z \quad \therefore \oint_C \left(\frac{\sin 2z}{z^4} \right) dz = \frac{2\pi i}{3!} f^{(3)}(0)$$

$$n+1=4 \Rightarrow n=3$$

$$z_0 = 0$$

$$f(z) = \sin z \quad f'(z) = \cos z \quad f''(z) = -\sin z \quad f'''(z) = -\cos z$$

$$f'''(0) = -\cos 0 = -1$$

$$\therefore \oint \frac{\sin z}{(z-0)^4} dz = \frac{2\pi i}{3!} f'''(0)$$

$$= \frac{2\pi i}{3!} \times -1 = -\frac{2\pi i}{3}$$

b) Find the Taylor series of $\frac{1}{1+z}$ about the centre $z_0 = i^0$.

Ans

$$f(z) = \frac{1}{1+z}$$

$$= \frac{1}{1+z+i^0-i^0}$$

$$= \frac{1}{1+i^0} \left[1 + \frac{z-i^0}{1+i^0} \right]^{-1}$$

$$= \frac{1}{1+i^0} \left[1 - \frac{z-i^0}{1+i^0} + \frac{(z-i^0)^2}{(1+i^0)^2} - \dots \right]$$

$$= \frac{1}{1+i^0} - \frac{(z-i^0)}{(1+i^0)^2} + \frac{(z-i^0)^2}{(1+i^0)^3} - \dots$$

18. a) Evaluate $\int_0^{1+i^0} (x-y+i^0 x^2) dz$ along the parabola $y=x^2$.

Ans $y=x^2 \Rightarrow dy=2x dx$.

$$z=x+iy \Rightarrow dz=dx+i dy.$$

$$=dx+i 2x dx$$

$$=(1+i 2x) dx \quad x \rightarrow 0 \text{ to } 1$$

$$\int_0^{1+i^0} (x-y+i^0 x^2) dz = \int_0^1 (x-x^2+i^0 x^2) (1+i 2x) dx$$

$$= \underline{\underline{-\frac{1}{3} + \frac{i^0}{2}}}$$

b) Evaluate $\oint_C \frac{\log z}{(z-4)^4} dz$ counter clockwise around the circle $|z-3|=2$.

Ans $\log z$ is analytic at all pts inside and on the circle $|z-3|=2$.

The singularity $z=4$ lie inside the circle $|z-3|=2$.

Apply Cauchy integral formula.

$$\oint_C \frac{\log z}{(z-4)^2} = \underline{\underline{\frac{\pi i^0}{2}}}$$

19) a) Find the Laurent's Series expansion of $\frac{z^2-1}{z^2-5z+6}$ about $z=0$ in the region $2 < |z| < 3$.

Ans

$$f(z) = \frac{z^2-1}{z^2-5z+6}$$

$$= \frac{z^2-1}{(z-2)(z-3)}$$

$$\begin{array}{r} 1 \text{ (Q)} \\ (z-2)(z-3) \overline{) \begin{array}{r} z^2-1 \\ -z^2+5z-6 \\ \hline 5z-7 \end{array}} \\ 5z-7 \text{ (R)} \end{array}$$

$$= Q + \frac{R}{D}$$

$$= 1 + \frac{5z-7}{z^2-5z+6}$$

$$= 1 + \left[\frac{-3}{z-2} + \frac{8}{z-3} \right] \text{ --- (1)}$$

$$\Rightarrow 2 < |z| < 3 \Rightarrow 2 < |z|, \quad |z| < 3$$

$$\Rightarrow \left| \frac{2}{z} \right| < 1 \quad \left| \frac{z}{3} \right| < 1$$

$$(1) \Rightarrow f(z) = 1 - \frac{3}{z(1-\frac{2}{z})} + \frac{8}{-3(\frac{z}{3}+1)}$$

$$= 1 - \frac{3}{z} \left[1 - \frac{2}{z} \right]^{-1} - \frac{8}{3} \left[1 - \frac{z}{3} \right]^{-1}$$

$$= 1 - \frac{3}{z} \left[1 + \left(\frac{2}{z} \right) + \left(\frac{2}{z} \right)^2 + \dots \right] - \frac{8}{3} \left[1 + \left(\frac{z}{3} \right) + \left(\frac{z}{3} \right)^2 + \dots \right]$$

b) Evaluate $\int_0^{2\pi} \frac{d\theta}{\sqrt{2} - \cos \theta}$

Ans $C: |z|=1$, put $z = e^{i\theta}$, $d\theta = \frac{dz}{iz}$, $\cos \theta = \frac{z+z^{-1}}{2}$.

$$\int_0^{2\pi} \frac{d\theta}{\sqrt{2} - \cos \theta} = 2i \oint_{C: |z|=1} \frac{dz}{z^2 - 2\sqrt{2}z + 1} \quad \text{--- (1)}$$

Poles are $z = \sqrt{2} + 1, \sqrt{2} - 1$

$\therefore \sqrt{2} - 1$ lies inside the region

$\therefore$ Residue at $\sqrt{2} - 1$ is $= \underline{\underline{-\frac{1}{2}}}$

Apply Residue theorem in (1)

$$\textcircled{1} \Rightarrow \int_0^{2\pi} \frac{d\theta}{\sqrt{2} - \cos \theta} = \underline{\underline{2\pi}}$$

20. a) Evaluate $\oint_C \frac{z-23}{z^2-4z-5} dz$ where

$C: |z-2-i| = 3.2$ using Residue theorem.

Ans $\oint_C \frac{z-23}{z^2-4z-5} dz$ where $C: |z-2-i| = 3.2$

Singular pts $\Rightarrow z^2 - 4z - 5 = 0$
 $\Rightarrow z = -1, 5$

$\therefore$ Residue of $f(z)$ at $z=5$ is -3

Residue of $f(z)$ at $z=-1$ is 4

$\therefore$ By Residue theorem.

$$\oint_C \frac{z-23}{z^2-4z-5} dz = 2\pi i [-3+4] \\ = \underline{\underline{2\pi i}}.$$

b) Evaluate $\int_0^\infty \frac{(x^2+2)dx}{(x^2+1)(x^2+4)}$

Ans

$$\int_0^\infty \frac{(x^2+2)dx}{(x^2+1)(x^2+4)} = \frac{1}{2} \int_{-\infty}^\infty \frac{(x^2+2)dx}{(x^2+1)(x^2+4)}$$

consider, $I = \int_C f(z) dz$, where $f(z) = \frac{z^2+2}{(z^2+1)(z^2+4)}$.

C is the path consisting of the semi circular path C_R along with diameter from $-R$ to $+R$.

$$\text{poles are } \Rightarrow (z^2+1)(z^2+4) = 0$$

$$\Rightarrow z = \pm i, z = \pm 2i$$

$z=i, z=2i$ are inside the region

$$\text{Res at } z=i \Rightarrow \frac{1}{6i} // \quad \text{Res at } z=2i \Rightarrow \frac{1}{6i} //$$

$\therefore$ By Res. theorem

$$\int_0^\infty \frac{(x^2+2)dx}{(x^2+1)(x^2+4)} = \underline{\underline{\frac{\pi}{3}}}.$$